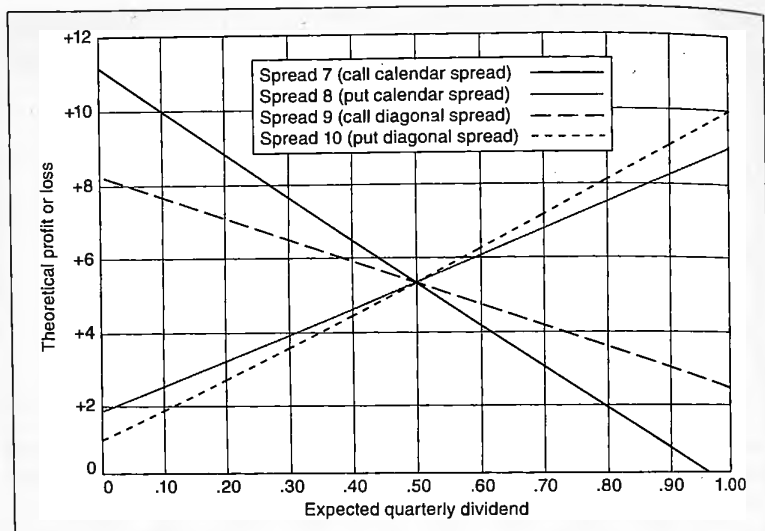


Figure 13-16 Dividend sensitivity.



gamma, the potential profit when the underlying market moves. The risk is the theta, the money that will be lost through the passage of time if the underlying market fails to make sufficiently large moves. The trader would like the reward (the gamma) to be as large as possible compared with the risk (the theta). We might express this relationship as a ratio

$$\text{gamma}/\text{theta}$$

The larger the absolute value of this ratio, the more efficient the position.

In the same way, a trader who has a negative gamma and a positive theta wants the risk (the gamma) to be as small as possible compared with the reward (the theta). He therefore wants the absolute value of the gamma/theta ratio to be as large as possible.

For example, we might go back and calculate the efficiency of Spreads 1 through 3 in Figure 13-3. The efficiencies are

|          | Gamma/Theta   | Efficiency |
|----------|---------------|------------|
| Spread 1 | -406.0/0.4235 | 959        |
| Spread 2 | -165.5/0.1365 | 1,212      |
| Spread 3 | -370.0/0.4000 | 925        |

Because each spread has a negative gamma and positive theta, we want the efficiency to be as small as possible. We can see that Spread 3 is best, which is consistent with our previous analysis of each spread.

Assuming that all strategies have approximately the same theoretical edge, the efficiency can be a reasonable method of quickly comparing strategies where all options expire at the same time. In such cases, the gamma and theta